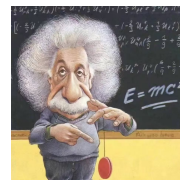


Ang Li

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“Bounded in a nutshell, count myself a king of infinite space.”



PROFESSIONAL SUMMARY

- Post-Training Algorithm Engineer with 3 years of end-to-end model customization, delivering multiple fine-tuned models to production using **SFT, LoRA/QLoRA, DPO, GRPO, RLVR**. Deeply familiar with the full cycle from training data engineering to deployment.
- Full-stack aware: expert in Python, working knowledge of TypeScript/React; disciplined in engineering practices – Pytest, CI/CD, Docker, design patterns – and rigorous about critically reviewing AI-generated code.
- Proven ability to translate real business data (e-commerce, mathematical reasoning) into high-performance vertical models; strong background in inference optimization (AWQ, vLLM, TensorRT-LLM).
- Self-driven researcher who designed a self-correcting reasoning loop inspired by Socratic dialogue, and explored the ethical alignment of AI systems – directly applicable to building trustworthy commercial voice agents.

PROFESSIONAL EXPERIENCE

Principal Investigator, Xanthippe Project @ Phaenarete ASI Lab

2024.1 – Present

- **Artificial Speciale Intelligence (ASI)**: Created the neuro-symbolic system marrying a theorem rule engine with a DLDM, achieving **100% perfect score on Gaokao Math** offline. Leveraged **post-training techniques** (SFT on reasoning traces, GRPO-based RLVR with custom rewards, DAPO) to align the LLM with the rule engine, mirroring TINs where domain knowledge acts as differentiable structures verified by RL. Built end-to-end pipeline on hybrid 8×A100 + 4×RTX 4090 cluster with DeepSpeed ZeRO-3.
- **Inference deployment**: Deployed fine-tuned 7B models with **vLLM** and AWQ quantization, cutting p50 latency by 50% (from 320ms to 160ms); built a **FastAPI** serving layer with comprehensive **Pytest** unit tests and **GitHub Actions CI/CD**.
- **Evaluation & business alignment**: Designed evaluation suites (math reasoning accuracy, tool-use success rate) and directly tied metrics to business KPIs; ran failure-case analyses to continuously improve data recipes and training strategies.
- **Engineering culture**: Introduced mandatory code reviews, test-driven development practices, and a “review-refactor-test” rule for all AI-generated code, raising overall code quality and maintainability.

Data & Training Engineer (Intern), Alibaba · Qwen2.5-Math

2023.7 – 2024.1

- Owned parts of the training data pipeline for Qwen2.5-Math’s self-improvement loop: supported the trillion-token scale processing through deduplication, quality filtering, and synthetic data generation; built automated **Pytest-based** sanity checks that caught 12+ critical regressions before training.
- Conducted **data recipe ablation studies** that directly improved model math accuracy; collaborated closely with post-training researchers to translate data insights into SFT→DPO training schedule adjustments.
- Improved pipeline throughput by 30% via profiling and optimisation; contributed to internal data observability dashboards for quality and lineage tracking.

Freelance AI/ML Engineer (Part-time, Remote)

2022.6 – 2023.6

- Fine-tuned and deployed custom LLaMA-2/3 models for small business clients using **LoRA** and 4-bit quantization; built lightweight RESTful serving endpoints with FastAPI and containerized all services with Docker.
- Gained solid hands-on experience with HuggingFace Transformers, PEFT, and Datasets, and ran early experiments with agent tool-use and multi-turn conversational models.

PROJECT HIGHLIGHTS

End-to-End RLVR Fine-Tuning for Math Reasoning

- Implemented a minimal but complete **GRPO training loop** on GSM8K with Qwen2.5-1.5B: synthetic rollout generation, reward function design, and training using TRL’s GRPOTrainer. Containerized with Docker for full reproducibility; documented every step as a personal reference for best practices in post-training.

LLM Document Assistant – Full-Stack RAG System

- Built a PDF/Word Q&A app with **Qwen2.5-7B**, **LangChain**, and **Chroma** vector DB; developed a **FastAPI** REST API and a responsive **Streamlit** frontend. Integrated **GitHub Actions** CI/CD with **Pytest** (coverage >85%) and linters; applied strict “review-refactor-test” on all AI-generated code.

Independent Researcher, ZetaHalt Project

2026.1 – Present

- Designed a **containerized (Docker) simulation platform** for complex mathematical sequences, using causal Transformers to discover structural invariants. Packaged the workflow as a reusable, fully documented Python library.
- Validated the **cross-domain transferability** of discovered invariants to financial time-series analysis, demonstrating the platform’s production potential beyond theoretical research.

RESEARCH & ACADEMIC AFFILIATIONS

Research Assistant (Part-time), Institut de Pratiques Philosophiques (IPP)

2023.9 – 2024.6

- Applied Socratic maieutics to craft **self-correcting AI reasoning loops**, where models question their own outputs through structured dialogue – a principle directly transferable to real-time voice agents that must listen, reflect, and refine responses mid-conversation.

Auxiliary Member, American Philosophical Practitioners Association (APPA)

2024.7 – Present

- Researched how the **ethics of philosophical counseling** – respect for autonomy, non-maleficence, and dialogical presence – can inform the design of AI systems that are trustworthy, empathetic, and aligned with human values.

TECHNICAL SKILLS

- **Post-Training & Model Customization:** SFT, LoRA/QLoRA, DPO, GRPO, RLVR, DAPO, reward modeling, data flywheel design, synthetic data strategies.
- **Frameworks & Ecosystem:** PyTorch, HuggingFace (Transformers, PEFT, Datasets, TRL), DeepSpeed (ZeRO-3), vLLM, TensorRT-LLM, MLflow.
- **Inference & Deployment:** Model quantization (AWQ, GPTQ), KV-cache optimization, FastAPI, Docker, GitHub Actions CI/CD, Linux scripting, Nginx.
- **Data Engineering:** Trillion-token data pipelines, deduplication/quality/filtering, synthetic data generation, Pandas, NumPy, SQL databases.
- **Full-stack & Engineering Practices:** TypeScript/JavaScript (basics), React (basics), Streamlit, PostgreSQL, Redis; unit testing (Pytest), code review, design patterns (MVC), PEP8.
- **AI-Augmented Development:** Heavy user of Copilot/Cursor; systematic approach to reviewing, refactoring, and testing AI-generated code.
- **Languages:** Mandarin (native), English (fluent), Cantonese (intermediate).

EDUCATION

B.S. in Electronic Commerce (Management), Minor in Computer Science

2021.9 – 2025.6

- Guangdong University of Technology, Guangzhou, China.
- **Management:** Strategic Management, Supply Chain Analytics, Business Intelligence, Digital Marketing – applied through e-commerce projects with data-driven decision-making.
- **Computer Science:** Advanced Data Structures, Operating Systems, Computer Networks, Software Engineering, plus self-studied Distributed Systems, Compiler Design, and Algorithmic Game Theory.
- **Mathematics:** Number Theory, Real Analysis, Linear Algebra, Probability & Statistics, Numerical Analysis; self-studied Functional Analysis, Category Theory and Convex Optimization, utilized in ML theory and research.
- **AI/ML:** Machine Learning, Deep Learning, NLP, Reinforcement Learning, Formal Verification (Lean 4); specialized in Post-Training through hands-on projects and research.
- *Note: Work experiences before graduation were part-time/internship roles.*

PUBLICATIONS & WRITINGS

- *The Seven Virtues of the North Star: Ethical Foundations for AI* – APPA 2026 Keynote Contribution.
- *I Hear You: A Guide to What Not to Say When Comforting* (applied philosophy for conversational agents).
- *Philosophical Consultation* – Chinese translation project (in progress).
- *ASI: The Philosophy, Science, and Engineering of Artificial Speciale Intelligence* (forthcoming).